

Week 1 - Class 2: Integers, Width, and Bool

Why This Matters

Integers in C++ have fixed width. Width determines the range. Unlike Python ints, they do not grow arbitrarily. We also introduce bool, the type for true/false values, which behaves differently than Python's bools.

Width and Ranges

Width = number of bits. More bits = more range.

Table:

8-bit: unsigned 0..255, signed -128..127

16-bit: 0..65535, -32768..32767

32-bit: 0..4 billion, -2 billion..+2 billion

64-bit: huge ranges.

Signed vs Unsigned

Unsigned uses all values for nonnegative. Signed uses half for negatives.

Two's complement represents negatives.

Bool Type

bool is its own type in C++. Values: true and false.

Size: typically 1 byte (not 1 bit).

Conversions: 0->>false, nonzero->>true.

Printing: default prints 0/1, use std::boolalpha for true/false.

```
#include <iostream>
using namespace std;
int main(){
    bool b1 = true;
    bool b2 = false;
    cout << b1 << " " << b2 << "\n"; // 1 0
    cout << boolalpha << b1 << " " << b2 << "\n"; // true false
}
```

Example Program

Print int sizes and ranges, plus bool behavior.

```
#include <iostream>
#include <limits>
using namespace std;
int main(){
    cout << "sizeof(int)="<<sizeof(int)<<" bytes\n";
    cout << "int min="<<numeric_limits<int>::min()<<" max="<<numeric_limits<int>::max()<<"\n";
    cout << "sizeof(bool)="<<sizeof(bool)<<" byte\n";
    cout << "bool values: "<<true<<" "<<false<<"\n";
}
```

Common Mistakes

1. Assuming int is always 32 bits.
2. Forgetting signed overflow is undefined behavior.
3. Assuming unsigned behaves like signed (it wraps silently).
4. Misunderstanding bool: it is usually 1 byte, not 1 bit.
5. Mixing bool and int without realizing true=1 and false=0.

Exercises

1. Print sizeof(bool).
2. Assign int to bool and print.

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3. Compare bool behavior in if statements.
4. Print min/max for int, short, long, long long.

Project Tie-In

Extend Type Explorer Project to:

- Print sizes and ranges for bool and int.
- Show int->bool conversions.
- Demonstrate signed vs unsigned overflow.